

Lab-3: 4-bit Ripple Carry Adder

Aim:

To design and verify a 4-bit ripple carry adder and to study the propagation delay on Logisim.

Theory:

A **4-bit Ripple Carry Adder (RCA)** is built by cascading four Full Adders. Each Full Adder receives the Carry-out of the previous stage as its Carry-in, causing the carry to "ripple" from the LSB to the MSB.

Propagation Delay is the time taken for a change in input to produce a stable output. In a Ripple Carry Adder, the carry must propagate through each Full Adder stage sequentially, so the total delay accumulates:

- Total Delay = $n \times t_{FA}$

where n = number of bits and t_{FA} = propagation delay of one Full Adder stage.

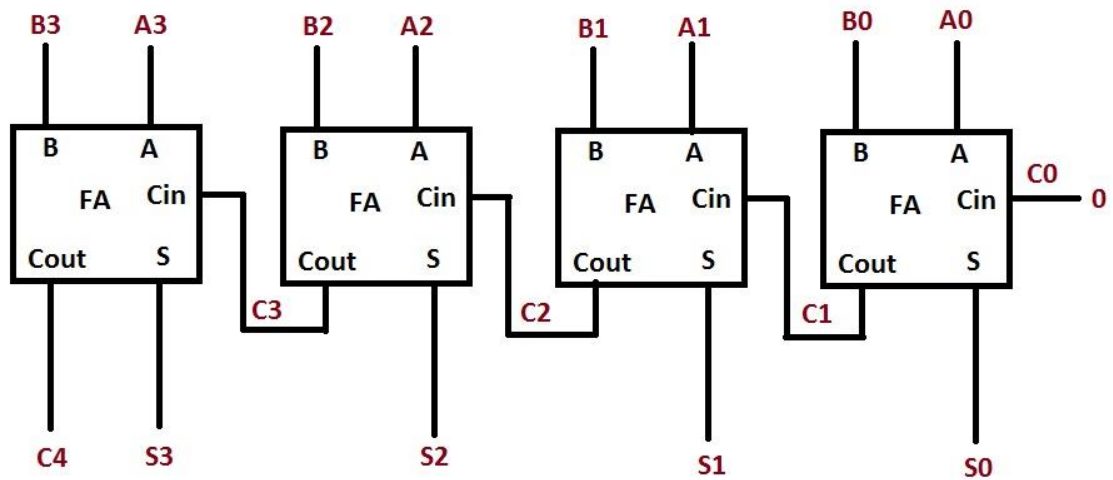
For a 4-bit RCA, the worst case occurs when carry ripples through all 4 stages (e.g., adding 0111 + 0001), making it slower than carry look-ahead adders for larger bit widths.

The **Chronogram** in Logisim is a timing diagram tool that displays how output signals change over time in response to input changes, allowing visual study of propagation delay.

Procedure:

1. Open Logisim and design a 4-bit Ripple Carry Adder by cascading four Full Adder modules.
2. Connect Cout of each stage to Cin of the next stage.
3. Add clock/pulse inputs to drive the input bits A and B.
4. Attach a **Chronogram** component to observe the Sum (S0–S3) and Cout signals over time.
5. Simulate the circuit and apply various input combinations using the clock.
6. Observe the timing diagram in the chronogram to identify the delay between input change and stable output.
7. Note the worst-case propagation delay when carry ripples across all 4 stages.

A 4 BIT RIPPLE CARRY ADDER



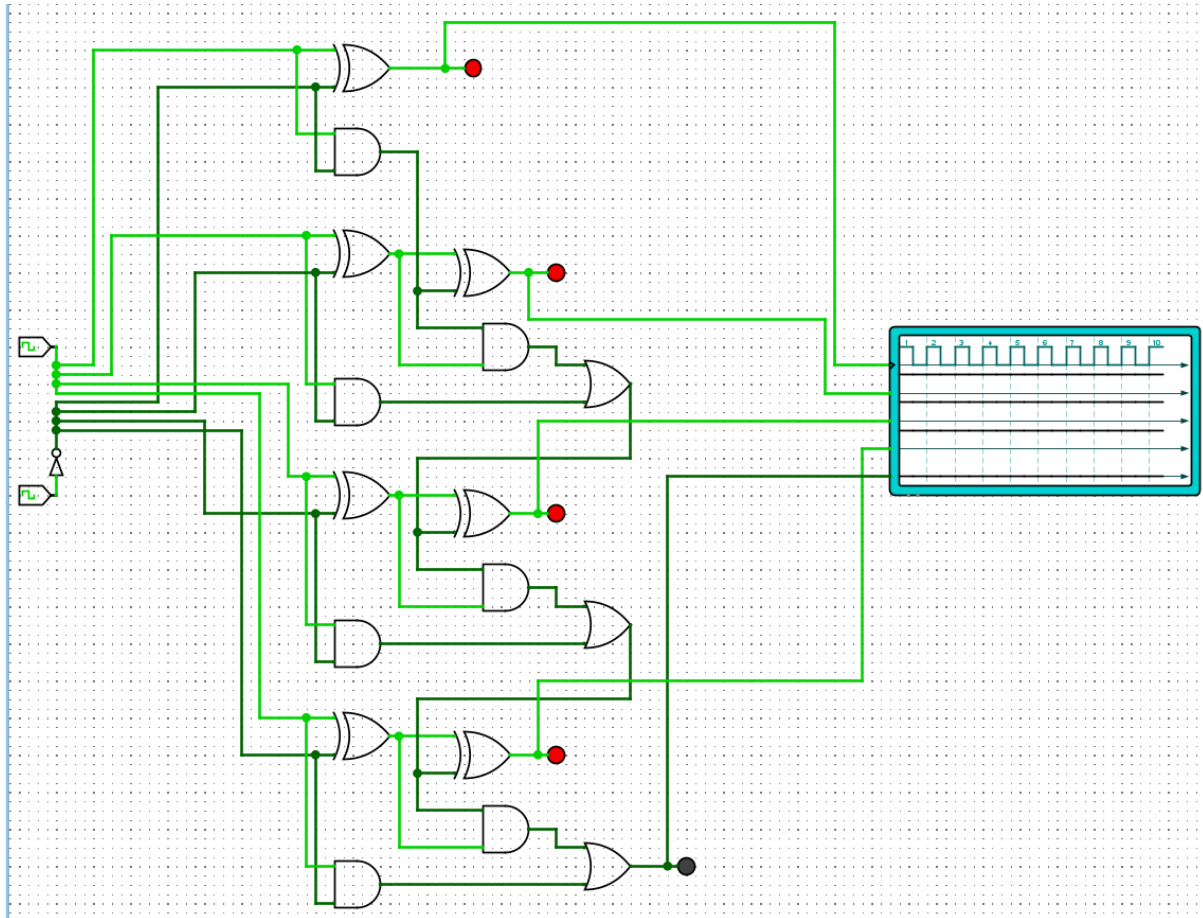
FA - Full Adders
B - Data 2
C - Carry

A - Data 1
S - Sum

Ripple Carry Adder-Truth Table

A ₄	A ₃	A ₂	A ₁	B ₄	B ₃	B ₂	B ₁	S ₄	S ₃	S ₂	S ₁	Carry
0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	1	0	0	1	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0	1
1	0	1	0	1	0	1	0	0	1	0	0	1
1	1	0	0	1	1	0	0	1	0	0	0	1
1	1	1	0	1	1	1	0	1	1	0	0	1
1	1	1	1	1	1	1	1	1	1	1	0	1

Circuit Diagram:



Conclusion:

The 4-bit Ripple Carry Adder was successfully designed and simulated in Logisim. Binary addition was verified for all tested input combinations. The chronogram confirmed that propagation delay increases with the number of stages the carry must ripple through — the worst case being when carry propagates across all four Full Adder stages. This experiment highlights the key limitation of RCA: increasing bit width linearly increases delay, making Carry Look-Ahead Adders (CLA) preferable for high-speed applications.